

Professional Summary

AI research engineer focused on LLM/VLM and computer vision with end-to-end delivery from model development to deployment. Experience includes RAG systems, OCR with vLLM, object detection and segmentation, and production APIs using FastAPI/Flask. Best Paper Award (ICNGC 2023); seeking AI roles in South Korea with impact on real-world products.

Education

- 2nd Mar-22 **Sejong University**, Seoul, South Korea
18th Feb-24
 - M.S. in Computer Science and Engineering
 - Thesis: FlameNet: An Efficient Deep Learning Framework for Fire Detection and Classification in Smart City Environment
 - GPA: 4.25/4.5, [Presentation Link](#)

1st Feb-14 **The Islamia University of Bahawalpur**, Bahawalpur, Pakistan
23rd Apr-18
 - B.S. in Computer Science
 - GPA: 3.28/4.0

Selected Publications

- Smart Cities **Visual Intelligence in Smart Cities: A Lightweight Deep Learning Model for Fire Detection in an IoT Environment**
IF 7.0, Q1
[Muhammad Nadeem](#), Naqqash Dilshad, Norah Saleh Alghamdi, L Minh Dang, Hyoung-Kyu Song, Junyoung Nam, Hyeonjoon Moon
[\[MDPI, PDF\]](#)
- Plants **VPBR: An Automatic and Low-Cost Vision-Based Biophysical Properties Recognition Pipeline for Pumpkin**
IF 4.5, Q1
L. Minh Dang, [Muhammad Nadeem](#), Tan N. Nguyen, Han Yong Park, O New Lee, Hyoung-Kyu Song, Hyeonjoon Moon
[\[MDPI, PDF\]](#)
- ITCCSCC-2023 **Machine Learning for Mental Health: A Systematic Study of Seven Approaches for Detecting Mental Disorders**
[Muhammad Nadeem](#), Junaid Rashid, Hyeonjoon Moon, Arailym Dosset
[\[The 38th International Technical Conference on Circuits/Systems, Computers and Communications, PDF\]](#)
- ICTC-2023 **Wheat Diseases Recognition Using Optimal Features Assisted Modified Soft Attention Network**
[Muhammad Nadeem](#), Aqib Khan, Ji-Won Kim, L. Minh Dang, Hyeonjoon Moon
[\[The 15th International Conference on ICT Convergence\]](#)
- ICNGC-2023 **Brest Cancer Recognition Through Visual Intelligence Assisted Lightweight Convolution Neural Network**
Best Paper Award
[Muhammad Nadeem](#), Haseeb Khan, Wisal Khan, L. Minh Dang, Nguyen Le Quan, Hyeonjoon Moon
[\[The 9th International Conference on Next Generation Computing \(ICNGC 2023\)\]](#)

Research and Work Experience

- 15th April-2024 **Computer Vision Researcher**, Spilab, Seoul, South Korea
Present **Tools:** LLM/VLM (RAG, LangChain, vLLM), CV (YOLOv8/12/26), Deployment (FastAPI, Flask), Edge (OpenVINO, TFLite), Tracking (Deep SORT)
- Foreign object debris detection using drone
 - Sea water waste detection using drone
 - Person and object tracking
 - Speech Emotion Recognition
 - Time-series forecasting with N-BEATS and N-HiTS for prediction tasks.
 - OCR pipelines with vLLM for real-world text.
 - Visual question answering prototypes combining vision backbones and LLMs.
 - Write research papers, technical reports, and present the work at conferences and meetings.
 - Model quantization and lightweight conversion.
 - Drone Based Fire and Smoke.
 - Multilingual AI Chatbot Development.
- 15th Mar-22 **Computer Vision Engineer**, Sejong University, Seoul, South Korea
28th Feb-24 **Tools:** Segmentation (Mask R-CNN), Generative (Stable Diffusion), Training/Fine-tuning, Dataset development, RNNs, GANs
- Working on Image Classification, Image Segmentation and Object Detections methods.
 - Worked on Data Parallel, Distributed Data-Parallel, and Multi-Node Training.
 - Training, fine-tuning CV models, transfer learning, and evaluating their performance.
 - Hands-on experience on recent object detection models (YOLO, FasterRCNN, etc.).
 - Write research papers, and technical reports, and present the work at conferences and meetings.
- 1st Jul-2018 **Software Engineer**, Systems Junction, Lahore, Pakistan
30th June-2021 **Tools:** Android (Java, Android Studio), Networking (Retrofit), Storage (Room), Media (Glide/Picasso), Backend (Firebase)
- Designing, developing, and maintaining Android applications using Java and XML programming languages.
 - Writing clean, efficient, and maintainable code that follows best practices and coding standards.
 - Troubleshooting and debugging issues, and implementing fixes to ensure smooth functioning.
 - Providing technical guidance and mentorship to junior team members, as needed.
 - Work on technical documentation, such as architectural diagrams, code, and user manuals.
 - Worked on MVC and singleton pattern, real-time push notifications, Firebase, realm
 - Optimizing computer vision algorithms for performance and memory usage on mobile devices.
 - Maintaining and improving existing CV codebases, including refactoring and bug fixing.
 - Developing and maintaining image processing pipelines and data processing workflows.

Honors and Achievements

- 2023 **Best Paper Award**, ICNGC 2023
2022 **Full Tuition Fee Waiver**, School of Computer Engineering, Sejong University
2022 **Prof. Stipend**, CVPR Lab, Sejong University
2023 **Conference Travel Grant**, Ministry of Education, South Korea
2014 **Merit Scholarship for Bachelor Degree**, The Islamia University of Bahawalpur, Pakistan

English Proficiency and Relevant Courseworks

- IELTS-AC 6.0** Listening 6.0, Speaking 6.0, Writing 6.0, Reading 5.5
Languages English (4/5), Korean (1/5)
A+ Introduction to Deep Learning, Advanced Deep Learning, Advanced AI
A+ Computer Vision Topics
A0 Advanced HCI

Technical Skills

- LLM/VLM** RAG, LangChain, vLLM, prompt engineering, VQA

Object Det	YOLOv8, YOLOv12, YOLOv26, SSD, YOLO-NAS, YOLOX, Faster R-CNN, DETR, GroundingDINO
Time-Series	Forecasting, prediction pipelines, N-BEATS, N-HiTS
Edge Deployment	Raspberry Pi 4/5, NVIDIA Jetson Orin Nano
DL techniques	Tracking, Optical Flow, Particle Filtering, Pose Estimation
CV Algorithms	Image Classification and Detection, Segmentation, Tracking
Soft. Control	JIRA, Agile, Scrum and GIT or Subversion
Lang	Python, Keras, JAVA, C++, C#
Editor	Visual Studio Code, PyCharm, Jupyter Notebook
OS	Windows 11, Ubuntu 24.04, macOS
APIs	PyTorch, TensorFlow, Caffe, OpenCV, FastAPI, Flask

References

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